Rule-Based Auto-Remediation for Windows Event Viewer Error Logs

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***Abstract*—Abstract- Modern enterprise computing infrastructures face significant operational challenges due to high volumes of system alerts and administrative event logs, which lead to severe alert fatigue and extended recovery latency. Traditional monitoring systems merely flag errors, leaving the diagnostic and recovery actions to manual administrator intervention. This paper introduces a lightweight, intelligent system for rule-based automated remediation within client and server operating system environments. The proposed architecture consists of a script-based event collector agent integrated with a centralized web service and a transactional database for execution audit trails. To prevent repetitive service restarts and handle complex system failures, we implement a multi-event inference mechanism using chronological event correlation with a five-minute sliding window. The engine enriches incoming log data with pre-configured catalog metadata, detects root cause variants via regular expression string matching, and scores confidence levels to select the appropriate recovery workflow. Additionally, we integrate a deep system repair fallback path executing system file checker and deployment image servicing utilities when critical operating system modules fail. For interactive operator control, a real-time web dashboard provides status tracking, system metrics, and manual approval gates for sensitive actions. Empirical evaluations demonstrate that the system reduces the average recovery latency for common administrative failures to under three seconds, mitigating alert volume by up to eighty-four percent through deduplication.**

***Index Terms*—Automated runbooks, chronological event correlation, operating system logging, self-healing systems.**

I Introduction

In modern enterprise computing operations, maintaining high availability and minimizing system downtime is of paramount importance [1]. A major source of diagnostic information for Windows-based client and server infrastructures is the Windows Event Log system, which records detailed information regarding system errors, service failures, security audits, and application crashes [2]. However, enterprise system administrators are frequently overwhelmed by thousands of alerts daily, a phenomenon known as alert fatigue [3]. Due to this high volume, critical system warnings and errors are often ignored or delayed in resolution, leading to service degradation and financial loss. Furthermore, the traditional incident management workflow relies heavily on manual intervention, where an administrator must manually log into the affected machine, examine the Event Viewer, determine the root cause, and execute recovery commands [4]. This manual process introduces significant recovery latency, increasing the overall Mean Time to Resolution (MTTR).

To address these limitations, several studies have focused on building automated runbook systems and self-healing IT infrastructures [5], [6]. Nonetheless, existing monitoring tools either lack integrated active remediation capabilities or are highly complex, expensive, and resource-heavy enterprise suites (such as Microsoft SCOM or Splunk SOAR) that are difficult to deploy in lightweight or localized environments. Additionally, standard automated triggers often suffer from a lack of context, executing simplistic actions (like restarting a service) repeatedly without identifying root causes, such as memory exhaustion or file system corruption, which leads to infinite failure-restart loops.

To overcome these challenges, we introduce an intelligent, lightweight, and end-to-end *Rule-Based Auto-Remediation System for Windows*. Our system comprises a lightweight PowerShell collector agent, a Flask REST API backend, and a dynamic web dashboard built with Flutter. By processing Windows Administrative Events (Errors and Warnings), our architecture enriches incoming events using a comprehensive metadata catalog and runs them through a multi-event chronological correlation engine to detect compound root causes. If system files are corrupted, the engine automatically escalates to a deep system repair fallback path using SFC and DISM.

Experimental evaluation of the system shows that it successfully captures and processes system alerts with near-zero latency, automatically resolves common administrative errors within seconds, and significantly reduces the alert volume via built-in deduplication mechanisms. The system maintains an interactive operator-in-the-loop workflow, allowing sensitive remediation scripts to go through a manual approval gate before execution, thus guaranteeing system security and stability.

I-A Motivation

Windows administrative systems produce vast streams of event logs, but traditional monitors only flag problems without taking corrective action [2]. System administrators must perform repetitive troubleshooting tasks, such as clearing disk space, restarting crashed services, or running system scans [4]. Such manual work is prone to delays, errors, and high operational costs. Furthermore, when applications crash due to corrupted OS files, simple app restarts fail repeatedly, highlighting the need for an intelligent system that detects root causes and escalates to system-wide healing automatically [5]. These operational challenges motivate the design of a lightweight, self-healing system that operates with high speed

and precision while maintaining an audit trail of all automated recovery actions.

I-B Contribution

The main contributions of this research are outlined as follows:

- Design of an end-to-end auto-remediation architecture combining a lightweight PowerShell collector, a Flask REST API backend, and a Flutter web dashboard.
- Development of a chronological event correlation engine that analyzes a 5-minute sliding window of events to identify compound root causes (e.g., service crashes caused by memory pool exhaustion) rather than isolated symptoms.
- Implementation of a deep system repair fallback mechanism that parses application crashes to detect core operating system failures and executes repair utilities automatically.
- Implementation of a priority-based rule execution engine with context parameter injection and operator-in-the-loop approval workflows for sensitive administrative tasks.

II Related Work

II-A IT Operations Automation and Monitoring

Enterprise log monitoring is the cornerstone of system reliability, providing the telemetry necessary to assess infrastructure health [6]. Standard event log infrastructures, such as Windows Event Forwarding (WEF) and Syslog, are widely used to centralize system logs for analysis [2]. While structured parsing and anomaly detection techniques [7] are highly effective at collecting and analyzing logs, they generally operate as passive monitoring tools. In contrast, runbook automation (RBA) compiles routine recovery procedures into automated workflows [5]. Modern orchestration platforms manage system drift, but they are not designed for reactive, event-driven active remediation. Event-driven automation platforms execute active remediation workflows, but they require substantial infrastructure overhead and are difficult to deploy on isolated Windows systems.

II-B Event Correlation and Deduplication

Diagnosing complex failures requires correlating multiple telemetry streams over time to discover the underlying root cause [8]. Autonomic computing research utilizes the Monitor-Analyze-Plan-Execute (MAPE) loop to enable self-managing properties in software systems [1]. However, a primary challenge in event-driven recovery is alert fatigue, where redundant notifications fire during a single incident [3]. Grouping events with identical signatures within a sliding time window helps suppress repeating alarms [9]. In this paper, we adapt chronological correlation and deduplication to Windows system events to suppress duplicate alerts and identify compound root causes (such as service crashes caused by memory pool exhaustion) before dispatching remediation.

II-C Interactive Self-Healing Systems

Fully autonomous recovery actions carry inherent security and operational risks, as misconfigured scripts can corrupt system states or disrupt critical services [4]. Human-in-the-loop (HITL) workflows address this concern by inserting manual authorization steps for high-risk operations [10]. Our work integrates log ingestion, chronological correlation, and active remediation in a single lightweight package. Unlike resource-heavy enterprise suites, our system runs locally, executing adaptive recovery procedures via PowerShell context injection while maintaining administrative approval gates for safety.

III Problem Formulation and Objectives

The primary objective of this research is to design and implement a self-healing system that monitors Windows Administrative Events (Errors and Warnings), enriches them with catalog metadata, detects root cause correlations, and dispatches priority-based remediation scripts.

Formally, let the event stream S be a sequence of events $S = (e_1, e_2, e_3, \ldots, e_n)$, where each event e_i is defined as a tuple $e_i = (t_i, ID_i, Src_i, Lvl_i, Msg_i, Host_i)$ representing the timestamp, Event ID, Source, Level (Error/Warning), Message string, and Hostname.

Let $D(e_i, W)$ be a deduplication function that returns 1 if there exists a prior event e_j in the event stream S sharing the same Event ID and hostname such that the timestamp difference $t_i - t_j < W$, and returns 0 otherwise. If $D(e_i, W) = 1$, the event is flagged as a duplicate and its active remediation trigger is suppressed to prevent redundant execution loops.

For unique events, the system applies a correlation function $(C_{cause}, P_{level}) = C(e_i, S_W)$ that inspects the historical event log $S_W = \{e_j \in S \mid 0 < t_i - t_j \leq W\}$ to identify co-occurring failures. For example, if the trigger event e_i represents a Service Control Manager crash (Event 7031) and the window contains a nonpaged pool allocation failure (Event 2019), the correlation engine returns $(C_{cause}, P_{level}) = (\text{memory exhaustion}, \text{high})$.

Finally, the system matches the event and its correlation context against a rule catalog R to dispatch a remediation action $A(e_i, C_{cause}) \rightarrow \text{ExecuteRemediation}(\text{Runner}, \text{Script}, \text{Parameters})$, where $Parameters$ includes the event details and correlation variables injected into the execution context.

The main objectives of this study are:

1) Development of a lightweight PowerShell event collector that subscribes to local event logs and forwards alerts to the backend with sub-second latency.
2) Construction of a rule matching and correlation engine that deduces compound causes from chronological log history and routes remediation to specific scripts.
3) Implementation of a secure execution runner that executes PowerShell remediation scripts with environment parameter injection and sandbox simulation support.
4) Design of an interactive Flutter web interface enabling real-time monitoring, manual approval gates, and con-

figuration of automated rules.
5) Validation of the system's healing efficiency, showing minimal CPU/Memory overhead and high success rates across common administrative errors.

IV System Architecture

The proposed system follows a client-server architecture split into three major layers: the PowerShell Event Collector, the Flask REST API Backend, and the Flutter Web Frontend. The architecture enables passive historical importing and active real-time event monitoring.

IV-A Lightweight PowerShell Event Collector

The PowerShell collector runs on host machines as a background scheduled task or system service. It uses WMI/CIM event queries to subscribe to the Windows Event Log service, listening specifically for Error (Level 2) and Warning (Level 3) entries in the System and Application channels. Upon detecting a new entry, it extracts the XML payload of the event, parses it into structured fields (Event ID, Source, Level, Message, TimeCreated, and Hostname), and transmits it as a JSON payload via an HTTP POST request to the backend API. The collector is designed to queue events locally if the API is temporarily unreachable, ensuring no loss of diagnostic telemetry.

IV-B Centralized Ingestion and Flask Web Service

The backend server is implemented using Flask and serves as the central API hub. It processes incoming JSON event payloads and manages the application state. Upon receiving an event, the API:

- Queries the SQLite database to perform deduplication checking.
- enriches the event by querying the event catalog database to append descriptive metadata (Category, Recommended Actions, Auto-remediate flags).
- Stores the enriched event in the SQLite database for audit logging.
- Passes the event to the Rules Engine for correlation and routing.

IV-C Rule Engine and Priority-Based Dispatcher

The Rules Engine is responsible for matching incoming events with configured remediation rules. Each rule specifies a trigger condition (such as Event ID and Source) and a target remediation script. The engine implements a priority-based dispatch queue, sorting execution requests based on event severity (Critical, High, Medium, Info) and correlation context. If a compound root cause is detected, the engine escalates the priority level and redirects the dispatch to a compound script designed to address the underlying issue first.

IV-D PowerShell Script Runner and Safe Execution

Remediation actions are executed by a dedicated execution runner that invokes PowerShell sub-processes. To pass event context dynamically without introducing shell-injection vulnerabilities, the runner injects event parameters into the script execution context as environment variables containing the event details. The runner supports a simulation mode where scripts execute dry-run checks and log their planned modifications to the database without altering the host configuration, allowing administrators to test rules safely.

IV-E Database Schema and Audit Trail

All application data is persisted in a local SQLite database using SQLAlchemy. The schema includes three core tables:

- **Events Table:** Records all ingested events, their enriched metadata, duplicate flags, and timestamps.
- **Rules Table:** Stores the remediation rules, matching criteria, execution script paths, and approval settings.
- **Remediation History Table:** Acts as a complete audit trail, logging execution timestamps, triggered rules, target hosts, console outputs, success status, and manual approval records.

V Proposed Method

V-A Event Collection and Log Parsing

The PowerShell collector utilizes a standard Windows event query cmdlet combined with an active query subscriber to read event records. The log parser extracts the dynamic data insertion strings from the event record to build a clean message representation. This structured parsing separates the static event description from dynamic components (like process names or disk paths), facilitating regular expression parsing.

V-B Event Ingestion and Deduplication Engine

When a new event e_i is received at the ingestion API endpoint, the backend checks for duplicates. A database query fetches the last event with the same ID and Host. If the difference in their timestamps is less than 5 minutes (300 seconds), the event's duplicate attribute is set to true, and it is logged without triggering any remediation rules. This deduplication process prevents "alert storms" from running identical recovery actions (e.g., multiple IIS restarts) during a single outage.

V-C Metadata Enrichment and Catalog Mapping

If an event is unique, the system enriches it using the preconfigured database catalog loaded from the event catalog. This catalog contains over 40 common Windows administrative errors. The mapping attaches:

- **Category:** E.g., Service Failure, Storage, Memory, System Integrity.
- **Severity:** Critical, High, Medium, or Info.
- **Recommended Action:** Human-readable description of standard resolution.

- **Auto-Remediate Candidate:** Boolean flag indicating if the error can be safely resolved automatically.

V-D Root Cause Variant Regex Analysis

Application crashes (Event 1000) are analyzed by the root cause analyzer using regular expressions. The parser extracts the "Faulting Application Path" and the "Faulting Module Name" from the event message. This allows the system to identify the specific executable that crashed and, more importantly, the DLL module that caused the exception (e.g., core system libraries).

V-E Chronological Multi-Event Correlation

The chronological correlation engine executes a lookup in the SQLite database for co-occurring events within a 5-minute sliding window. The engine matches findings against the correlation map to identify compound root causes. If a correlation is found, the system switches the remediation plan from a simple symptom-based response to a compound remediation script. Table I lists the primary multi-event correlation scenarios supported by the engine.

TABLE I
CHRONOLOGICAL MULTI-EVENT CORRELATION SCENARIOS

Trigger Event	Co-occurring Event(s)	Compound Cause
7031 (Service Crash)	2019/2020 (Memory Pool Exhaustion)	Memory exhaustion
55 (NTFS Corruption)	11 (Disk Controller), 51 (Paging Error)	Disk I/O error
1000 (App Crash)	8003/8004 (AppLocker Block)	Application control block
7000 (Service Start fail)	5157 (Firewall Packet Block)	Firewall service block
1101 (Audit Logs Drop)	2013 (Low Disk Space)	Disk space exhaustion

V-F System Repair Escalation Fallback

If the root cause analyzer detects that an application crash (Event 1000) was caused by a core operating system module, the system triggers the system repair fallback script. The repair executes in two phases:

- **Phase 1 (System File Check):** Runs a system file checker scan to verify and repair system files.
- **Phase 2 (System Image Restoration):** If the system file checker reports unrecoverable errors, the script executes the deployment image servicing tool to restore system image integrity.
- **Reboot Scheduling:** Schedules a system reboot with a 30-second warning to apply file changes.

V-G Priority Dispatch and Remediation Run

When a remediation script is triggered, the Flask backend invokes a PowerShell subprocess, passing context variables via environment variable injection. Scripts check the simulation mode configuration; if enabled, they log their planned actions to the database and exit, avoiding real system changes. For sensitive actions, the rule requires manual approval, pausing the script execution and rendering an "Approve" button on the web dashboard.

Algorithm 1 Auto-Remediation Event Processing

Input: Event payload $e = (t, ID, Src, Msg, Host)$
Output: Remediation status
if $D(e, 300) = 1$ **then**
  Log event as duplicate and **Exit**
end if
Query metadata catalog and enrich event e
$(C_{cause}, P_{level}) \leftarrow$ Correlate(e, S_W)
if $ID = 1000$ and core OS crash detected **then**
  $(C_{cause}, P_{level}) \leftarrow$ (core OS corruption, critical)
end if
Rule $R \leftarrow$ MatchRule(ID, Src, C_{cause})
if R is found **then**
  if R requires manual approval **then**
    Set status $\leftarrow$ Pending Approval; Await operator
  else
    Set context parameters with event details
    Execute remediation script and log history
  end if
else
  Log event as unhandled; Set status $\leftarrow$ No Rule Matched
end if

V-H Algorithm Summary

Algorithm 1 details the step-by-step logic executed by the Flask backend upon receiving a new event from the PowerShell collector.

VI Experimental Setup

VI-A Test Environment and Simulation Platform

To evaluate the auto-remediation system, we set up a dedicated Windows 11 Enterprise virtual machine (Intel Core i7-11700K @ 3.6GHz, 16GB RAM) running the Flask API backend and the local PowerShell collector. The SQLite database was initialized with the pre-configured metadata catalog containing 42 event definitions. To validate real-world remediation workflows, we developed a Python simulation script that writes authentic Windows error events directly into the local Event Log database using the Windows APIs or parses custom trigger logs.

VI-B Collector and Event Trigger Setup

The PowerShell event triggers were registered on the host system using an automated setup utility. A total of 27 Task Scheduler tasks were created under the automated task path. Each task is bound to a specific Event ID and Source (e.g., Event 7031 for Service Control Manager), configured to instantly execute the command-line event processor entry-point when an event matches the filter. This simulates a live, production event forwarder.

VI-C Baseline Resolution Frameworks

To assess the improvements in system healing efficiency, the proposed auto-remediation system was compared against two baseline workflows:

- **Manual Troubleshooting Baseline:** Simulates a standard sysadmin response workflow, where the event is logged,

an administrator is alerted, and they manually log in, diagnose the issue, and execute command-line fixes.
- **Standard Scripted Runbook Baseline:** Executes simple, event-triggered scripts that run immediately upon event detection without performing deduplication or checking chronological correlation.

VI-D Remediation Performance Metrics

The system's performance was evaluated using the following metrics:

- **Ingestion Latency:** Time elapsed from the generation of the Windows event log to its arrival at the Flask API.
- **Mean Time to Resolution (MTTR):** Time elapsed from event generation to the successful completion of the remediation script.
- **Deduplication Efficiency:** Percentage of alerts suppressed under high-frequency event storms.
- **Remediation Success Rate:** Percentage of test events successfully resolved on the host.
- **Resource Overhead:** CPU and memory usage of the background collector and API server.

VI-E System Specifications and Platform Setup

The Flask backend was run using Python 3.11 with SQLAlchemy ORM, and the Flutter web application was compiled to release-mode HTML/JS hosted on a local web server. Database queries were optimized using indexing on event IDs, hostnames, and timestamp columns to ensure minimal lookup latency during correlation execution.

VII Results and Discussion

VII-A Quantitative Performance and Scalability

Operational performance evaluations show that the auto-remediation system dramatically reduces system recovery latency. Standard manual troubleshooting took an average of 420 seconds (7 minutes) to resolve service crashes and disk space issues. The scripted runbooks resolved errors in 12 seconds, but failed during cascading issues (such as trying to restart a service on a disk with corrupted files). The proposed auto-remediation system achieved an average MTTR of **2.85 seconds** across all auto-remediable events, including script runner spawning, correlation lookup, and command execution. In addition, the event ingestion latency averaged 118 milliseconds over 500 test events, and the system remained robust under high volumetrics (buffering and processing 1,000 simultaneous requests without database corruption). Table II summarizes the performance comparison across the three resolution methods.

TABLE II
RESOLUTION PERFORMANCE COMPARISON

Resolution method	MTTR (s)	Deduplication (%)	Success rate (%)	CPU (%)
Manual Troubleshooting	420.00	0.0	98.5	N/A
Scripted Runbook	12.00	0.0	68.2	1.2
Proposed System	2.85	84.2	95.8	0.4

These results prove that the system provides near-instantaneous healing capabilities while reducing CPU utilization, thanks to the lightweight PowerShell WMI triggers.

VII-B Deduplication and Correlation Verification

During a simulated "alert storm" where a crashing service generated 120 service failure events (ID 7031) within 3 minutes, the deduplication engine successfully suppressed 119 duplicate execution triggers. Only the first event spawned the remediation script, reducing the host's processing overhead and preventing concurrent runner instances from locking system files. The total alert volume logged was reduced by 84.2%, mitigating operator alert fatigue. To test the chronological correlation engine, we simulated a memory leak that triggered Event 2019 (Nonpaged pool allocation failure) followed by Event 7031 (Service crash). The engine achieved 100% correlation accuracy, correctly routing the trigger to the memory exhaustion remediation script in all 50 test iterations, whereas the scripted runbook baseline repeatedly attempted to restart the service and failed due to out-of-memory errors.

VII-C System Repair Fallbacks and Safety Gates

The system repair fallback path was tested by triggering Event 1000 (Application Crash) with a core system module listed as the faulting module. Out of 20 test runs with simulated system file corruption, the system successfully repaired the OS files in 18 runs, reporting a 90% success rate. The two failures occurred when network access was disabled, preventing the image restoration utility from contacting the update servers. For sensitive operations, such as restarting critical services or clearing volume tables, manual approval gates successfully paused the script runner, logged a pending approval state, and resumed execution within 450 milliseconds once the operator clicked the approve button on the web dashboard, ensuring operational safety.

VII-D Discussion

The experimental results demonstrate that the rule-based auto-remediation system provides rapid, self-healing capabilities with minimal system overhead. Integrating chronological correlation resolves the root causes of service failures (like memory exhaustion or disk controller errors) rather than repeating symptomatic restarts. The deep system repair fallback path offers a graduated mechanism to handle core file corruption, which is typically unaddressable by standard runbooks.

However, the system's reliance on pre-configured rules means it cannot resolve uncataloged anomalies. If a new service crashes due to an unknown logic error, the system logs the event as unhandled. Thus, maintaining and updating the rule database is necessary to preserve the healing capabilities over time.

VIII Limitations

Despite the positive results demonstrated by the auto-remediation system, several limitations exist:

VIII-A Windows OS and Registry Dependence

The system is specifically designed for the Windows operating system, relying on Windows Event Logs, WMI/CIM event queries, and PowerShell scripts. It cannot monitor or remediate Linux or macOS servers, limiting its usage in heterogeneous data centers.

VIII-B Security Contexts and Administrative Privileges

To restart system services, execute system file checker and image restoration scans, and terminate processes, the PowerShell runner must execute under the local administrator or system security context. This requirement increases the attack surface if the Flask API backend or the database is compromised, necessitating strict network isolation.

VIII-C Rule Base Maintenance and Scaling

The engine is dependent on static, pre-defined rules. As the enterprise infrastructure scales and new applications are deployed, system administrators must manually write and maintain matching rules and PowerShell remediation scripts. It does not automatically adapt to changing environments.

VIII-D Network and Database Latency under Log Storms

Under severe event storms generating thousands of events per second, the synchronous write operations to the local SQLite database may introduce processing latency, potentially delaying remediation triggers.

VIII-E Scope of Automated Healing Actions

The system is restricted to scripting actions that can be executed locally on the host. It cannot resolve complex, distributed hardware failures or physical network disconnections that require physical intervention.

IX Future Work

To improve the scalability and capabilities of the auto-remediation system, the following future enhancements are proposed:

- **Cross-Platform Expansion:** Develop a lightweight Python collector agent for Linux systems to parse system logs and execute Bash remediation scripts.
- **LLM-Based Script Generation:** Integrate a local Large Language Model (LLM) to automatically generate custom PowerShell remediation scripts based on the event description and error message for uncataloged events.
- **Enterprise SIEM/SOAR Connectors:** Build native integration links to forward correlated event sequences to enterprise systems like Splunk, Azure Sentinel, or ServiceNow.
- **Anomalous Event Rate Detection:** Implement machine learning-based anomaly detection models to flag irregular surges in specific event categories, enabling proactive healing before critical errors occur.
- **Distributed Authentication:** Implement secure OAuth2 authentication and end-to-end payload encryption between remote event collectors and the centralized backend API.

X Conclusion

This paper presents an intelligent, lightweight, and end-to-end *Rule-Based Auto-Remediation System for Windows*. By combining a PowerShell event collector with a Flask REST API backend and a Flutter web dashboard, the system automates the detection, diagnosis, and remediation of common Windows administrative errors.

The implementation of chronological event correlation allows the engine to analyze a 5-minute sliding window of events, identifying compound root causes (like memory pool exhaustion or disk paging errors) and resolving them before attempting symptomatic service restarts. Furthermore, the root cause analyzer parses application crashes to detect core OS DLL corruption, automatically executing SFC and DISM system repairs.

Experimental results confirm that the system achieves an average MTTR of 2.85 seconds, compared to 7 minutes for manual troubleshooting, while reducing logged alert volume by 84.2% through deduplication. The system operates with negligible CPU and memory overhead, offering a robust, secure, and scalable self-healing solution that reduces operational costs and maintains high infrastructure availability.

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